



Signals and Systems EE252, Spring 2026

Instructor: Tariq Mumtaz

Semester week # 13





Course Learning Outcome (CLOs):

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)



Roadmap till Semester end:

Frequency Domain Analysis of CT/DT Signals & Systems:

Z- Transform (Wk 7 – 11)

➔ Laplace Transform (Wk 12 – 13)

Fourier Transform (Wk 14 – 15)

Project presentations (Wk16)



Assessment Roadmap till Semester end:

Assessment Method	Percentage
Quizzes (4-5)	20%
Assignments	10%
Midterm Examination	25%
Course Project (Complex Engineering Problem)	15%
Final Examination	30%

	Weight (%)	Total	Completed	Remaining	Topic	Due Week
Quiz	20	5	4	2	Laplace and Fourier	14, 15
Homework	10	4	3	1	Laplace and Fourier	15
Midterm Exam	25	1	1			
Project M1	7.5	1	1			
Project M2	7.5	1		1		16
Final Exam	30	1		1	All CLOs	
Total	100					

Quiz: CP is worth one quiz credit.
Quiz: N-1 (best five out of 6)



Learning outcome of Wk#12 session:

The time-domain analysis of linear, time-invariant, continuous-time (LTIC) systems.

Time Domain Analysis (Chapter 02)

- CT System modeling by Differential equations
- CT signal and system representations
- CT system impulse response $h(t)$
- CT Convolution- $y(t) = x(t) * h(t)$
- Properties of CT systems using $h(t)$:
 - Causality
 - BIBO Stability

Frequency Domain Analysis (Chapter 04)

- Eigenfunction of CT systems
- The Laplace Transform



Learning outcome of Wk#13 session:

Frequency Domain Analysis (Chapter 04)

- Eigenfunction of CT systems
- The Laplace Transform
- The Laplace Transform Properties
- Inverse s – transform $X(s)$ to $x(t)$ – Partial fractions expansion
- CT system modeling- Differential equations (time domain) to algebraic equations (s -domain)
- CT system Total response – Zero Input response & Zero state response
- CT system Transfer function $H(s)$ – Poles/ Zeros & BIBO Stability Check



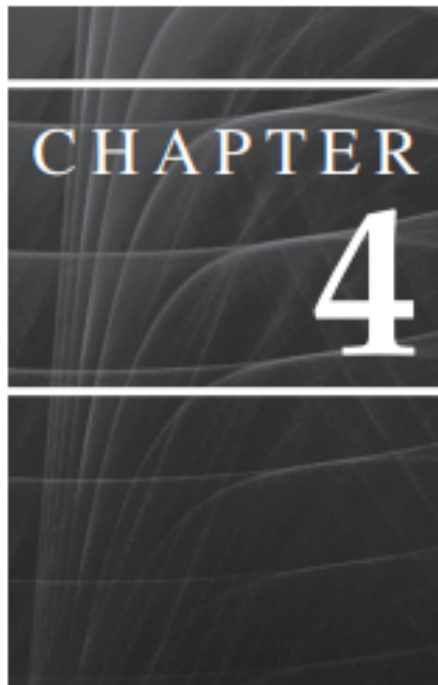
LTIC vs LTID systems :

A quick comparison with LTI CT and DT systems:

- Notion of time changes from n to t , i.e., $y[n] \rightarrow y(t)$
- The summation operation $\sum x[n]$ changes to Integration $\int x(t)$
- The difference equations change to Differentiation equations to model the CT system
- Eigenfunction Z^n ($z = \gamma e^{j\Omega}$) changes to e^{st} ($s = \sigma + j\omega$)



Frequency Domain Analysis of LTIC Systems:



CONTINUOUS-TIME SYSTEM ANALYSIS USING THE LAPLACE TRANSFORM



Frequency Domain analysis- LTIC Systems:

For a CT signal $x(t)$, its Laplace Transform $X(s)$ is defined as

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$X(s)$ exist for only those values of S where the integral converges.

Here, $s = \sigma + j\omega$

$$x(t) \Longleftrightarrow X(s)$$

$$e^{st} = e^{(\sigma + j\omega)t}$$

$$e^{st} = e^{\sigma t} e^{j\omega t}$$

$e^{\sigma t} \rightarrow$ represents a growing/decaying factor

$e^{j\omega t} \rightarrow$ represents the frequency of the sinusoid

Useful Signals Models:

The exponential function e^{st}

The complex frequency “ s ” is represented on a complex frequency plane (s plane). The s plane is separated into two parts:

- The left half-plane (LHP) corresponds to exponentially decaying signals
- The right half-plane (RHP) corresponds to exponentially growing signals.

The imaginary axis separates the two regions and corresponds to signals of constant amplitude.

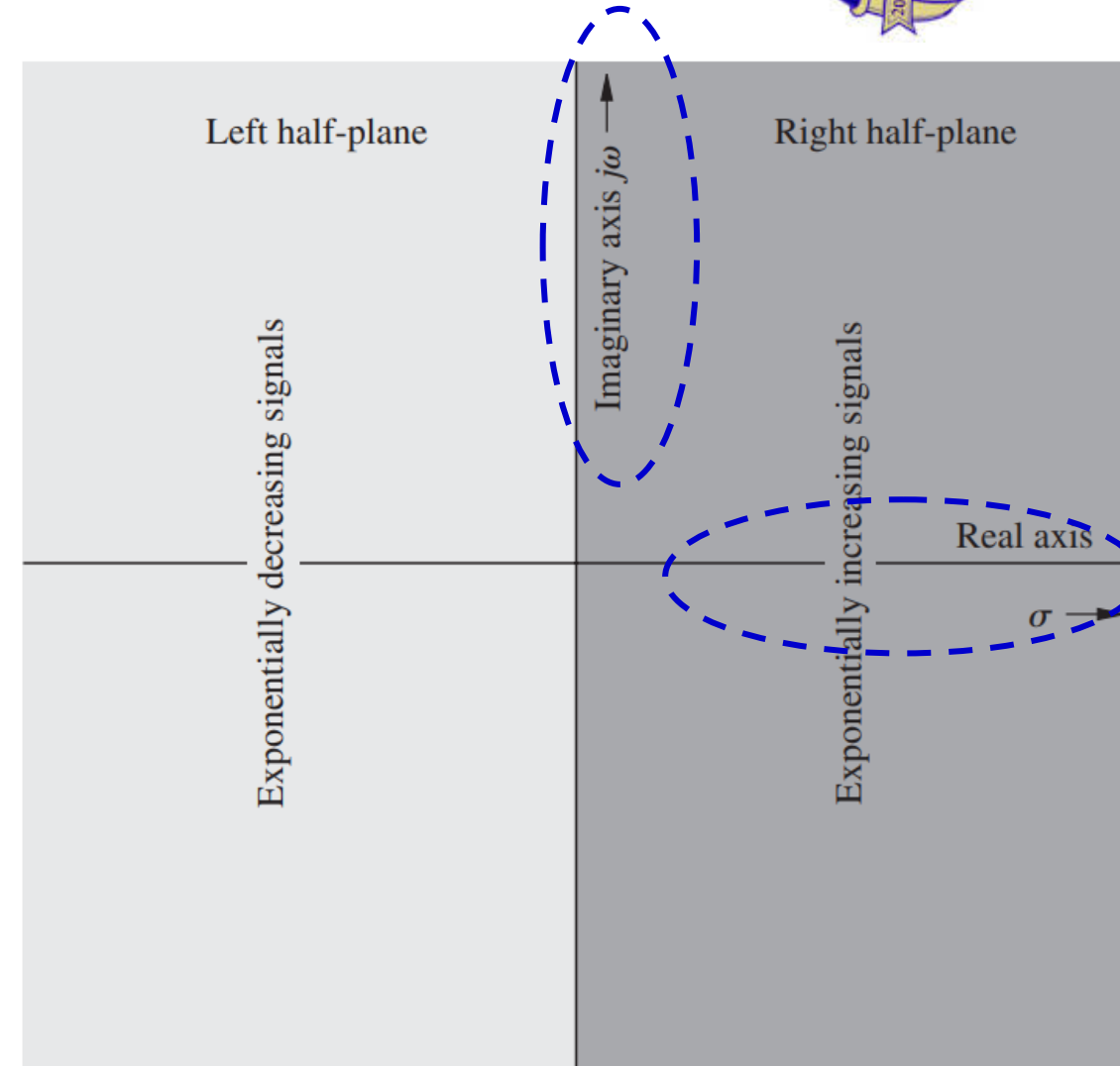


Figure 1.22 Complex frequency plane.



Frequency Domain analysis- LTIC Systems:

Bilateral Laplace Transform.

$$X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt$$

Unilateral Laplace Transform.

$$X(s) = \int_0^{\infty} x(t) e^{-st} dt$$

For Causal Signals

Inverse Laplace Transform.

$$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s) e^{st} ds$$

c is any real constant such that the contour lies in the Region of Convergence (ROC)

The Laplace Transform of Basic Signals

$X(s)$ and ROC

Frequency Domain analysis- LTIC Systems:

For a signal $x(t) = e^{-at}u(t)$, find the Laplace transform $X(s)$ and its ROC.

By definition,

$$X(s) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st} dt$$

Because $u(t) = 0$ for $t < 0$ and $u(t) = 1$ for $t \geq 0$,

$$X(s) = \int_0^{\infty} e^{-at}e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^{\infty}$$

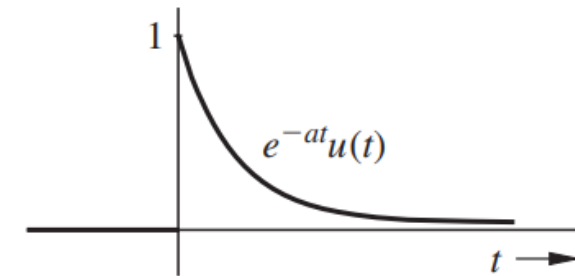
$$= \frac{1}{-(s+a)} [(e^{-(s+a)\cdot\infty}) - (e^{-(s+a)\cdot 0})]$$

$$\lim_{t \rightarrow \infty} e^{-(s+a)t} = \begin{cases} 0 & \text{Re}(s+a) > 0 \\ \infty & \text{Re}(s+a) < 0 \end{cases}$$

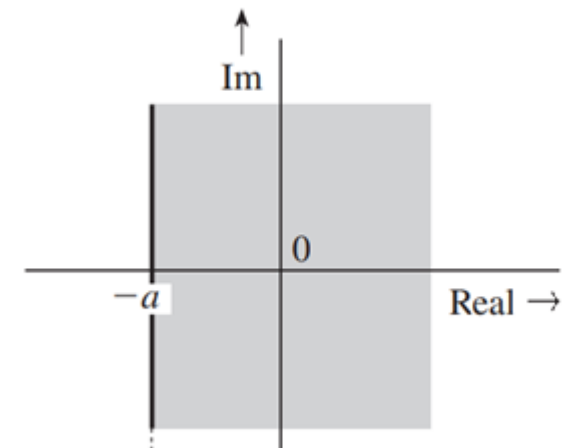
$$X(s) = \frac{1}{s+a} \quad \text{Re}(s+a) > 0$$

$$e^{-at}u(t) \iff \frac{1}{s+a} \quad \text{Re } s > -a$$

Signal $x(t)$



Region of convergence



Frequency Domain analysis- LTIC Systems:

$$x(t) = -e^{-at}u(-t)$$

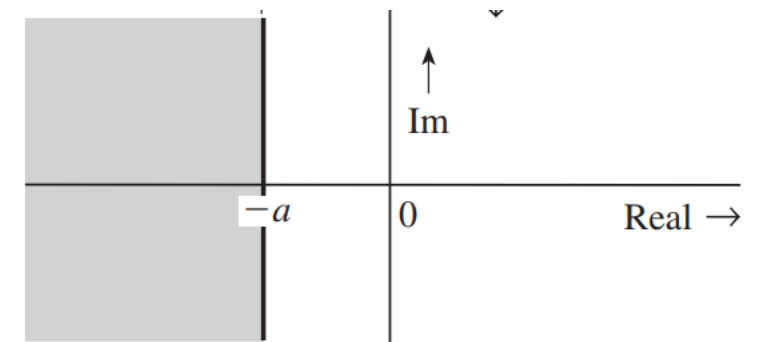
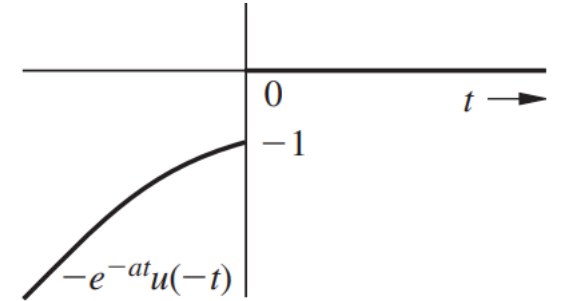
$$X(s) = \int_{-\infty}^{\infty} -e^{-at}u(-t)e^{-st} dt$$

Because $u(-t) = 1$ for $t < 0$ and $u(-t) = 0$ for $t > 0$,

$$X(s) = \int_{-\infty}^0 -e^{-at}e^{-st} dt = - \int_{-\infty}^0 e^{-(s+a)t} dt = \frac{1}{s+a} e^{-(s+a)t} \Big|_{-\infty}^0$$

$$\lim_{t \rightarrow -\infty} e^{-(s+a)t} = 0 \quad \text{Re}(s+a) < 0$$

$$X(s) = \frac{1}{s+a} \quad \text{Re } s < -a$$



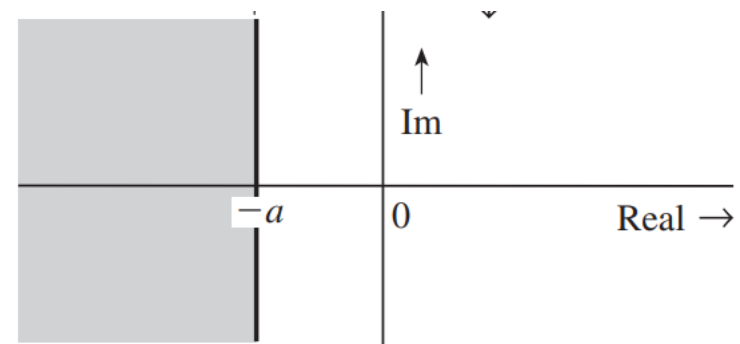
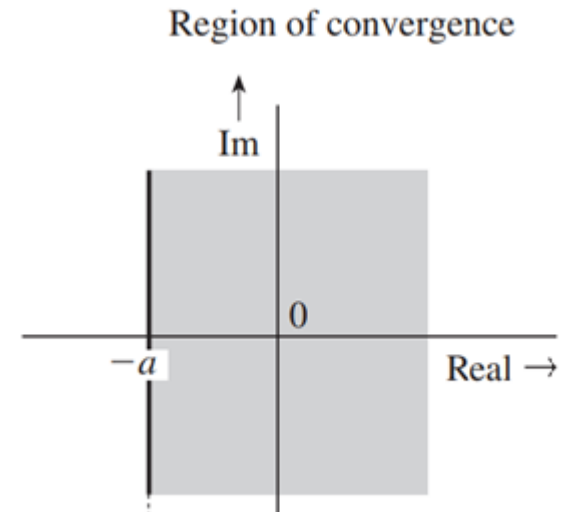
Frequency Domain analysis- LTIC Systems:

- The following two functions have the same Laplace transform $X(s)$ but entirely different ROCs.

$$x(t) = e^{at}u(t)$$

$$x(t) = -e^{at}u(-t)$$

- This implies that the transform function alone is not sufficient to describe the function in the Laplace domain. The union of the ROCs of the above two functions is the complete s -plane





Frequency Domain analysis- LTIC Systems:

$$x(t) = \delta(t)$$

$$X(s) = \int_0^{\infty} \delta(t) e^{-st} dt$$

$$\int_{-\infty}^{\infty} \phi(t) \delta(t - \tau) dt = \phi(\tau)$$

$$X(s) = e^{-s \cdot 0}$$

$$X(s) = 1$$

ROC = Entire S Plane



Frequency Domain analysis- LTIC Systems:

$$x(t) = \delta(t - t_0)$$

$$t_0 > 0$$

$$X(s) = \int_0^{\infty} \delta(t - t_0) e^{-st} dt$$

$$\int_{-\infty}^{\infty} \phi(t) \delta(t - \tau) dt = \phi(\tau)$$

$$X(s) = e^{-st_0}$$

ROC = Entire S Plane



Frequency Domain analysis- LTIC Systems:

$$x(t) = u(t)$$

Since $u(t) = 1$ for $t \geq 0$:

$$X(s) = \int_0^{\infty} u(t) e^{-st} dt$$

$$X(s) = \int_0^{\infty} e^{-st} dt$$

$$X(s) = \frac{1}{s} e^{-st} \bigg|_0^{\infty}$$

$e^{-st} \rightarrow 0$ as $t \rightarrow \infty$ only if $\text{Re}(s) > 0$

$$X(s) = \frac{1}{s}$$

ROC: $\text{Re } s > 0$

Frequency Domain analysis- LTIC Systems:

$$x(t) = \cos(\omega_0 t)u(t) = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})u(t).$$

$$\begin{aligned}\int_{-\infty}^{\infty} e^{j\omega_0 t} u(t) e^{-st} dt &= \int_0^{\infty} e^{j\omega_0 t} e^{-st} dt = \int_0^{\infty} e^{-(s-j\omega_0)t} dt = \frac{1}{-(s-j\omega_0)} e^{-(s-j\omega_0)t} \Big|_0^{\infty} \\ &= \frac{1}{-(s-j\omega_0)} [(e^{-(s-j\omega_0)\cdot\infty}) - (e^{-(s-j\omega_0)\cdot 0})] = \frac{1}{-(s-j\omega_0)} (0 - 1) = \frac{1}{(s-j\omega_0)}, \text{Re}\{s\} > 0\end{aligned}$$

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-j\omega_0 t} u(t) e^{-st} dt &= \int_0^{\infty} e^{-j\omega_0 t} e^{-st} dt = \int_0^{\infty} e^{-(s+j\omega_0)t} dt = \frac{1}{-(s+j\omega_0)} e^{-(s+j\omega_0)t} \Big|_0^{\infty} \\ &= \frac{1}{-(s+j\omega_0)} [(e^{-(s+j\omega_0)\cdot\infty}) - (e^{-(s+j\omega_0)\cdot 0})] = \frac{1}{-(s+j\omega_0)} (0 - 1) = \frac{1}{(s+j\omega_0)}, \text{Re}\{s\} > 0\end{aligned}$$

$$\mathcal{L}\{x(t)\} = \frac{1}{2} \left[\frac{1}{(s-j\omega_0)} + \frac{1}{(s+j\omega_0)} \right] = \frac{s}{s^2 + \omega_0^2}, \text{Re}\{s\} > 0$$



Frequency Domain analysis- LTIC Systems:

Laplace Transform pair of
Basic signals

TABLE 4.1 Select (Unilateral) Laplace Transform Pairs

No.	$x(t)$	$X(s)$
1	$\delta(t)$	1
2	$u(t)$	$\frac{1}{s}$
3	$tu(t)$	$\frac{1}{s^2}$
4	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5	$e^{\lambda t} u(t)$	$\frac{1}{s - \lambda}$
6	$te^{\lambda t} u(t)$	$\frac{1}{(s - \lambda)^2}$
7	$t^n e^{\lambda t} u(t)$	$\frac{n!}{(s - \lambda)^{n+1}}$
8a	$\cos bt u(t)$	$\frac{s}{s^2 + b^2}$
8b	$\sin bt u(t)$	$\frac{b}{s^2 + b^2}$
9a	$e^{-at} \cos bt u(t)$	$\frac{s + a}{(s + a)^2 + b^2}$

Properties of the Laplace Transform



Frequency Domain analysis- LTIC Systems:

TABLE 4.2 Unilateral Laplace Transform Properties

Important points – key difference:

$x(0)$
 $x(0^-)$
 $x(0^+)$

Operation	$x(t)$	$X(s)$
Addition	$x_1(t) + x_2(t)$	$X_1(s) + X_2(s)$
Scalar multiplication	$kx(t)$	$kX(s)$
Time differentiation	$\frac{dx(t)}{dt}$	$sX(s) - x(0^-)$
	$\frac{d^2x(t)}{dt^2}$	$s^2X(s) - sx(0^-) - \dot{x}(0^-)$
	$\frac{d^3x(t)}{dt^3}$	$s^3X(s) - s^2x(0^-) - s\dot{x}(0^-) - \ddot{x}(0^-)$
	$\frac{d^nx(t)}{dt^n}$	$s^nX(s) - \sum_{k=1}^n s^{n-k}x^{(k-1)}(0^-)$
Time integration	$\int_{0^-}^t x(\tau) d\tau$	$\frac{1}{s}X(s)$
	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{s}X(s) + \frac{1}{s} \int_{-\infty}^{0^-} x(t) dt$
Time shifting	$x(t - t_0)u(t - t_0)$	$X(s)e^{-st_0}$ $t_0 \geq 0$
Frequency shifting	$x(t)e^{s_0t}$	$X(s - s_0)$
Frequency differentiation	$-tx(t)$	$\frac{dX(s)}{ds}$
Frequency integration	$\frac{x(t)}{t}$	$\int_s^{\infty} X(z) dz$
Scaling	$x(at), a \geq 0$	$\frac{1}{a}X\left(\frac{s}{a}\right)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi j}X_1(s) * X_2(s)$
Initial value	$x(0^+)$	$\lim_{s \rightarrow \infty} sX(s) \quad (n > m)$
Final value	$x(\infty)$	$\lim_{s \rightarrow 0} sX(s) \quad [\text{poles of } sX(s) \text{ in LHP}]$

$x(0^-)$: Represents initial condition

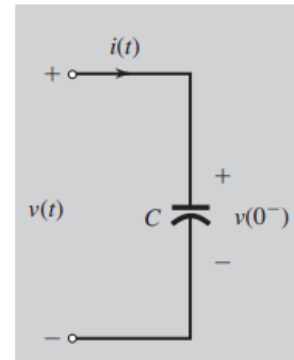
Frequency Domain analysis- LTIC Systems:

Representation of the initial condition

Consider a capacitor C with an initial voltage $v(0^-)$. The equation $i(t) = C \frac{dv(t)}{dt}$ holds.

Now take Laplace transform on both sides

$$I(s) = C(sV(s) - v(0^-)).$$



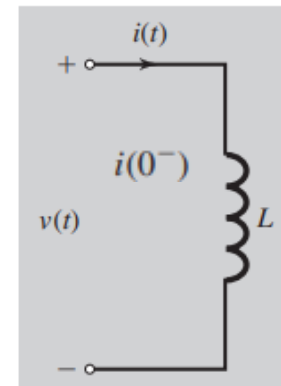
Time differentiation

$$\frac{dx(t)}{dt} \quad sX(s) - x(0^-)$$

Similarly, consider an inductor L with an initial current $i(0^-)$. The equation $v(t) = L \frac{di(t)}{dt}$ holds.

Now take Laplace transform on both sides

$$V(s) = L(sI(s) - i(0^-)) = LsI(s) - Li(0^-)$$



Frequency Domain analysis- LTIC Systems:

Proof: Time Shifting Property

Consider a causal signal $x(t)$ with Laplace transform $X(s)$. Suppose that we delay $x(t)$ by t_0 units of time to obtain $x(t - t_0)$ with $t_0 \geq 0$. The new signal will have Laplace transform

$$X(s) = \mathcal{L}\{x(t - t_0)\} = \int_0^{\infty} x(t - t_0)e^{-st} dt$$

Let $t - t_0 = v$. In that case $dt = dv$ and

$$\begin{aligned}\mathcal{L}\{x(t - t_0)\} &= \int_{-t_0}^{\infty} x(v)e^{-s(v+t_0)} dv = e^{-st_0} \int_{-t_0}^{\infty} x(v)e^{-sv} dv \\ &= e^{-st_0} \int_0^{\infty} x(v)e^{-sv} dv = e^{-st_0} X(s)\end{aligned}$$



Frequency Domain analysis- LTIC Systems:

Proof: Time Differentiation Property

Given that $\mathcal{L}\{x(t)\} = X(s)$ show $\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = sX(s) - x(0^-)$

$$\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

Partial integration states that: $\int_a^b u dv = uv|_a^b - \int_a^b v du$.

Using partial integration with $u = e^{-st} \Rightarrow du = -se^{-st} dt$ and $v = x(t) \Rightarrow dv = \frac{dx(t)}{dt} dt$

we have:

$$\begin{aligned} \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt &= \int_{0^-}^{\infty} e^{-st} d(x(t)) = e^{-st} x(t) \Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} x(t) (-s) e^{-st} dt = \\ &[e^{-s\infty} x(\infty) - e^{-s0^-} x(0^-)] + s \int_{0^-}^{\infty} x(t) e^{-st} dt \end{aligned}$$

The first term in the brackets goes to zero (as long as $x(t)$ doesn't grow faster than an exponential which is a condition for the existence of the transform). Therefore,

$$\mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt = -e^{-s0^-} x(0^-) + s \int_{0^-}^{\infty} x(t) e^{-st} dt = sX(s) - x(0^-).$$



Frequency Domain analysis- LTIC Systems:

Proof: Time Differentiation Property

Repeated application of the differentiation property yields the Laplace transform of the higher order derivatives of a signal.

$$\mathcal{L}\left\{\frac{d^2x(t)}{dt^2}\right\} = s[sX(s) - x(0^-)] - \dot{x}(0^-) = s^2X(s) - sx(0^-) - \dot{x}(0^-)$$
$$\vdots$$

$$\mathcal{L}\left\{\frac{d^nx(t)}{dt^n}\right\} = s^nX(s) - s^{n-1}x(0^-) - s^{n-2}\dot{x}(0^-) - \dots - x^{(n-1)}(0^-)$$

$$\text{with } x^{(r)}(0^-) = \left.\frac{d^rx(t)}{dt^r}\right|_{t=0^-}$$



Frequency Domain analysis- LTIC Systems:

Proof: Time Convolution Property

$$y(t) = x_1(t) * x_2(t)$$

$$\mathcal{L}\{x_1(t) * x_2(t)\} = X_1(s)X_2(s)$$

$$Y(s) = \int_{-\infty}^{\infty} y(t)e^{-st}dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau \right] e^{-st}dt$$

$$Y(s) = \int_{-\infty}^{\infty} x_1(\tau) \left[\int_{-\infty}^{\infty} x_2(t-\tau)e^{-st}dt \right] d\tau$$

$$\lambda = t - \tau \implies t = \lambda + \tau, \quad dt = d\lambda$$

$$\int_{-\infty}^{\infty} x_2(t-\tau)e^{-st}dt = \int_{-\infty}^{\infty} x_2(\lambda)e^{-s(\lambda+\tau)}d\lambda = e^{-s\tau} \int_{-\infty}^{\infty} x_2(\lambda)e^{-s\lambda}d\lambda = e^{-s\tau} X_2(s)$$

$$Y(s) = \int_{-\infty}^{\infty} x_1(\tau) [e^{-s\tau} X_2(s)] d\tau$$

$$= X_2(s) \int_{-\infty}^{\infty} x_1(\tau)e^{-s\tau}d\tau :$$

$$= X_1(s)X_2(s)$$



Frequency Domain analysis- LTIC Systems:

Proof: Time Convolution Property

Problem

Determine the convolution $c(t) = e^{at}u(t) * e^{bt}u(t)$.

Solution

We use the Laplace transforms:

$$\mathcal{L}\{e^{at}u(t)\} = \frac{1}{(s-a)} \text{ and } \mathcal{L}\{e^{bt}u(t)\} = \frac{1}{(s-b)}$$

$$\text{Therefore, } \mathcal{L}\{e^{at}u(t) * e^{bt}u(t)\} = \frac{1}{(s-a)(s-b)}.$$

$$\text{We see that } \frac{1}{(s-a)(s-b)} = \frac{1}{(a-b)} \left[\frac{1}{s-a} - \frac{1}{s-b} \right].$$

$$\text{Therefore, } \mathcal{L}^{-1} \left\{ \frac{1}{(s-a)(s-b)} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{(a-b)} \left[\frac{1}{s-a} - \frac{1}{s-b} \right] \right\} = \frac{1}{(a-b)}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s-a} - \frac{1}{s-b} \right\} \Rightarrow$$

$$c(t) = e^{at}u(t) * e^{bt}u(t) = \frac{1}{(a-b)} (e^{at} - e^{bt})u(t)$$



Frequency Domain analysis- LTIC Systems:

Proof: Initial and Final Value Theorem

Problem: How to find the initial and final values of a function $x(t)$ ($t \rightarrow 0^+$ and $t \rightarrow \infty$) if we know its Laplace Transform $X(s)$ and do not want to compute the inverse?

- **Initial Value Theorem**

$$\lim_{t \rightarrow 0^+} x(t) = x(0^+) = \lim_{s \rightarrow \infty} sX(s), \quad x(t) \text{ must be bounded on } (0, \infty).$$

Conditions

- Laplace transforms of $x(t)$ and $\frac{dx(t)}{dt}$ must exist.
- The power of the numerator of $X(s)$ is less than the power of its denominator.

- **Final Value Theorem**

$$\lim_{t \rightarrow \infty} x(t) = x(\infty) = \lim_{s \rightarrow 0} sX(s), \quad x(t) \text{ must be bounded on } (0, \infty).$$

Conditions

- Laplace transforms of $x(t)$ and $\frac{dx(t)}{dt}$ must exist.
- The poles of $sX(s)$ are all on the left plane or origin.



Frequency Domain analysis- LTIC Systems:

Proof: Initial and Final Value Theorem

Example: Find the initial and final values of $y(t)$ if $Y(s)$ is given by:

$$Y(s) = \frac{10(2s + 3)}{s(s^2 + 2s + 5)}$$

Initial Value

$$y(0^+) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} s \frac{10(2s+3)}{s(s^2+2s+5)} = \lim_{s \rightarrow \infty} \frac{10(2s+3)}{(s^2+2s+5)} = 0$$

Final Value

$$y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s \frac{10(2s+3)}{s(s^2+2s+5)} = \lim_{s \rightarrow 0} \frac{10(2s+3)}{(s^2+2s+5)} = 6$$

End of session # 01



Learning outcome of Wk#13 session:

Frequency Domain Analysis (Chapter 04)

- Eigenfunction of CT systems
- The Laplace Transform
- The Laplace Transform Properties
- Inverse s – transform $X(s)$ to $x(t)$ – Partial fractions expansion
- CT system modeling- Differential equations (time domain) to algebraic equations (s -domain)
- CT system Total response – Zero Input response & Zero state response
- CT system Transfer function $H(s)$ – Poles/ Zeros & BIBO Stability Check

Inverse of Laplace Transform using PFE



Frequency Domain analysis- LTIC Systems:

Practice problem 1: Find $x(t)$

$$X(s) = \frac{7s - 6}{s^2 - 5s - 6}$$

Factorize the denominator:

$$s^2 - s - 6 = (s + 2)(s - 3)$$

$$X(s) = \frac{K_1}{s + 2} + \frac{K_2}{s - 3}$$

$$K_1 = \left. \frac{7s - 6}{s - 3} \right|_{s=-2} = \frac{7(-2) - 6}{-2 - 3} = \frac{-14 - 6}{-5} = \frac{-20}{-5} = 4$$

$$K_2 = \left. \frac{7s - 6}{s + 2} \right|_{s=3} = \frac{21 - 6}{5} = \frac{15}{5} = 3$$



Frequency Domain analysis- LTIC Systems:

$$X(s) = \frac{K_1}{s+2} + \frac{K_2}{s-3}$$

$$X(s) = \frac{4}{s+2} + \frac{3}{s-3}$$

Inverse Laplace:

$$\frac{1}{s+a} \rightarrow e^{-at}u(t)$$

$$x(t) = 4e^{-2t}u(t) + 3e^{3t}u(t)$$



Frequency Domain analysis- LTIC Systems:

Practice problem 2: Find $x(t)$

$$X(s) = \frac{5e^{-2s}}{(s+1)(s+2)}$$

First, ignore the delay factor e^{-2s}

$$\begin{aligned} X'(s) &= \frac{5}{(s+1)(s+2)} \\ &= \frac{A}{s+1} + \frac{B}{s+2} \end{aligned}$$

At $s = -1$:

$$A = \frac{5}{-1+2} = 5$$

At $s = -2$:

$$B = \frac{5}{-2+1} = -5$$

$$X'(s) = \frac{5}{s+1} - \frac{5}{s+2}$$

Inverse Laplace:

$$x'(t) = 5e^{-t} - 5e^{-2t}$$

$$e^{-as}X(s) \longleftrightarrow x(t-a)u(t-a)$$

Apply shifting property:

Here, $a=2$ for the given delay factor e^{-2s}

$$x(t) = x'(t-2)u(t-2)$$

$$x(t) = [5e^{-(t-2)} - 5e^{-2(t-2)}] u(t-2)$$



Frequency Domain analysis- LTIC Systems:

Practice problem 3: Find $x(t)$

$$X(s) = \frac{2s^2 + 5}{s^2 + 3s + 2}$$

Partial fractions only work when the numerator degree < the denominator degree (proper fraction).
Then, what is the process in case:

$$\deg(N) \geq \deg(D)$$

Using long division, we obtain $H(s) = R(s) + H'(s)$, where $R(s)$ is an $(M - N)^{\text{th}}$ -order polynomial and $H'(s)$ is a proper transfer function.

$$\frac{2s^2 + 5}{s^2 + 3s + 2} = 2 + \frac{-6s + 1}{(s + 1)(s + 2)}$$

$$\begin{array}{r} 2 \\ \hline s^2 + 3s + 2 \quad \overline{) \quad 2s^2 + 5} \\ \underline{2s^2 + 4s + 6s} \\ -6s + 1 \end{array}$$



Frequency Domain analysis- LTIC Systems:

$$\frac{-6s + 1}{(s + 1)(s + 2)} = \frac{A}{s + 1} + \frac{B}{s + 2}$$

$$A = \left. \frac{-6s + 1}{s + 2} \right|_{s=-1} = \frac{-6(-1) + 1}{-1 + 2} = \frac{6 + 1}{1} = 7$$

$$B = \left. \frac{-6s + 1}{s + 1} \right|_{s=-2} = \frac{-6(-2) + 1}{-2 + 1} = \frac{12 + 1}{-1} = -13$$

$$X(s) = 2 + \frac{7}{s + 1} - \frac{13}{s + 2}$$

$$\begin{aligned} 2 &\rightarrow 2\delta(t) \\ \frac{1}{s+a} &\rightarrow e^{-at}u(t) \end{aligned}$$

$$x(t) = 2\delta(t) + 7e^{-t}u(t) - 13e^{-2t}u(t)$$



Frequency Domain analysis- LTIC Systems:

Read Annexure B.5

Characteristic equations under different conditions, including conjugate roots.

B.5 PARTIAL FRACTION EXPANSION

In the analysis of linear time-invariant systems, we encounter functions that are ratios of two polynomials in a certain variable, say, x . Such functions are known as *rational functions*. A rational function $F(x)$ can be expressed as

$$F(x) = \frac{b_mx^m + b_{m-1}x^{m-1} + \dots + b_1x + b_0}{x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0} = \frac{P(x)}{Q(x)} \quad (\text{B.22})$$

The function $F(x)$ is *improper* if $m \geq n$ and *proper* if $m < n$.[†] An improper function can always be separated into the sum of a polynomial in x and a proper function. Consider, for example, the function

$$F(x) = \frac{2x^3 + 9x^2 + 11x + 2}{x^2 + 4x + 3}$$

Because this is an improper function, we divide the numerator by the denominator until the remainder has a lower degree than the denominator.

$$\begin{array}{r} x^2 + 4x + 3 \overline{) 2x^3 + 9x^2 + 11x + 2} \\ \underline{2x^3 + 8x^2 + 6x} \\ x^2 + 5x + 2 \\ \underline{x^2 + 4x + 3} \\ x - 1 \end{array}$$

CT System Modeling

Differential Equations (time-domain) to Algebraic Equations (S- domain)

CT System Analysis: Total Response
Zero Input response + Zero State response

Frequency Domain analysis- LTIC Systems:

The LTIC system is modeled as differential equation (linear constant-coefficient differential equation (LCCDE):

$$\begin{aligned} \frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \dots + a_{N-1} \frac{dy(t)}{dt} + a_N y(t) \\ = b_{N-M} \frac{d^M x(t)}{dt^M} + b_{N-M+1} \frac{d^{M-1} x(t)}{dt^{M-1}} + \dots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t) \end{aligned}$$

where all the coefficients a_i and b_i are constants (due to Time-invariance). Using operator notation D to represent d/dt , we can express this equation as

$$\begin{aligned} (D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) y(t) \\ = (b_{N-M} D^M + b_{N-M+1} D^{M-1} + \dots + b_{N-1} D + b_N) x(t) \end{aligned}$$

Frequency Domain analysis- LTIC Systems:

The time-differentiation property of the Laplace transform has set the stage for solving linear differential (or integro-differential) equations with constant coefficients.

	$x(t)$	$X(s)$
Addition	$x_1(t) + x_2(t)$	$X_1(s) + X_2(s)$
Scalar multiplication	$kx(t)$	$kX(s)$
Time differentiation	$\frac{dx}{dt}$	$sX(s) - x(0^-)$
	$\frac{d^2x}{dt^2}$	$s^2X(s) - sx(0^-) - \dot{x}(0^-)$
	$\frac{d^3x}{dt^3}$	$s^3X(s) - s^2x(0^-) - s\dot{x}(0^-) - \ddot{x}(0^-)$
	$\frac{d^n x}{dt^n}$	$s^n X(s) - \sum_{k=1}^n s^{n-k} x^{(k-1)}(0^-)$
Time integration	$\int_{0^-}^t x(\tau) d\tau$	$\frac{1}{s}X(s)$
	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{s}X(s) + \frac{1}{s} \int_{-\infty}^{0^-} x(t) dt$

Frequency Domain analysis- LTIC Systems:

Practice Problem:

Problem: Solve the following second-order linear differential equation:

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

given that $y(0^-) = 2$, $\dot{y}(0^-) = 1$ and input $x(t) = e^{-4t}u(t)$.

Individual component in the time domain

$$\frac{dy(t)}{dt}$$

$$\frac{d^2y(t)}{dt^2}$$

$$x(t) = e^{-4t}u(t)$$

$$\frac{dx(t)}{dt}$$

Respective S- domain equivalent

$$sY(s) - y(0^-) = sY(s) - \mathbf{2}$$

$$s^2Y(s) - sy(0^-) - \dot{y}(0^-) = s^2Y(s) - \mathbf{2s - 1}$$

$$X(s) = \frac{1}{s + 4}$$

$$sX(s) - x(0^-) = \frac{s}{s + 4} - 0 = \frac{s}{s + 4}$$



Frequency Domain analysis- LTIC Systems:

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

$$(s^2Y(s) - 2s - 1) + 5(sY(s) - 2) + 6Y(s) = \frac{s}{s+4} + \frac{1}{s+4}$$

$$(s^2 + 5s + 6)Y(s) - (2s + 11) = \frac{s+1}{s+4}$$

$$\begin{aligned}(s^2 + 5s + 6)Y(s) &= \frac{s+1}{s+4} + (2s + 11) \\ &= \frac{s+1 + 2s^2 + 8s + 11s + 44}{s+4} \\ &= \frac{2s^2 + 20s + 45}{s+4}\end{aligned}$$

$$Y(s) = \frac{2s^2 + 20s + 45}{(s+2)(s+3)(s+4)}$$



Frequency Domain analysis- LTIC Systems:

$$Y(s) = \frac{2s^2 + 20s + 45}{(s + 2)(s + 3)(s + 4)}$$

$$Y(s) = \frac{13/2}{s + 2} - \frac{3}{s + 3} - \frac{3/2}{s + 4} \quad \text{PFE}$$

Inverse Laplace Transform:

$$y(t) = \left(\frac{13}{2} e^{-2t} - 3e^{-3t} - \frac{3}{2} e^{-4t} \right) u(t)$$



Frequency Domain analysis- LTIC Systems:

Zero Input & Zero State Response:

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

$$(s^2 + 5s + 6)Y(s) = \frac{s + 1}{s + 4} + (2s + 11)$$

$$Y(s) = \underbrace{\frac{2s + 11}{s^2 + 5s + 6}}_{\text{ZIR}} + \frac{s + 1}{(s + 4)\underbrace{(s^2 + 5s + 6)}_{\text{ZSR}}}$$

ZIR

Taking input zero

ZSR

Taking the initial condition zero

Frequency Domain analysis- LTIC Systems:

$$Y(s) = \frac{2s + 11}{s^2 + 5s + 6} + \frac{s + 1}{(s + 4)(s^2 + 5s + 6)}$$

$$Y(s) = \left(\frac{7}{s + 2} - \frac{5}{s + 3} \right) + \left(\frac{-\frac{1}{2}}{s + 2} + \frac{2}{s + 3} - \frac{\frac{3}{2}}{s + 4} \right)$$

$$y(t) = \underbrace{(7e^{-2t} - 5e^{-3t})u(t)}_{\text{ZIR}} + \underbrace{\left(-\frac{1}{2}e^{-2t} + 2e^{-3t} - \frac{3}{2}e^{-4t}\right)u(t)}_{\text{ZSR}}$$

ZIR

Taking input zero

ZSR

Taking the initial condition zero



Frequency Domain analysis- LTIC Systems:

DRILL 4.7 Laplace Transform to Solve a Second-Order Linear Differential Equation

Solve

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = 2\frac{dx(t)}{dt} + x(t)$$

for the input $x(t) = u(t)$. The initial conditions are $y(0^-) = 1$ and $\dot{y}(0^-) = 2$.

ANSWER

$$y(t) = \frac{1}{3}(1 + 9e^{-t} - 7e^{-3t})u(t)$$

	$x(t)$	$X(s)$
Addition	$x_1(t) + x_2(t)$	$X_1(s) + X_2(s)$
Scalar multiplication	$kx(t)$	$kX(s)$
Time differentiation	$\frac{dx}{dt}$	$sX(s) - x(0^-)$
	$\frac{d^2x}{dt^2}$	$s^2X(s) - sx(0^-) - \dot{x}(0^-)$
	$\frac{d^3x}{dt^3}$	$s^3X(s) - s^2x(0^-) - s\dot{x}(0^-) - \ddot{x}(0^-)$
	$\frac{d^nx}{dt^n}$	$s^nX(s) - \sum_{k=1}^n s^{n-k}x^{(k-1)}(0^-)$
Time integration	$\int_{0^-}^t x(\tau) d\tau$	$\frac{1}{s}X(s)$
	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{1}{s}X(s) + \frac{1}{s} \int_{-\infty}^{0^-} x(t) dt$

CT System Analysis

$H(s)$: The Transfer Function of a LTIC System

Frequency Domain analysis- LTIC Systems:

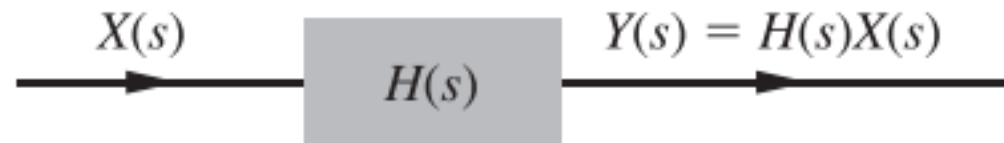
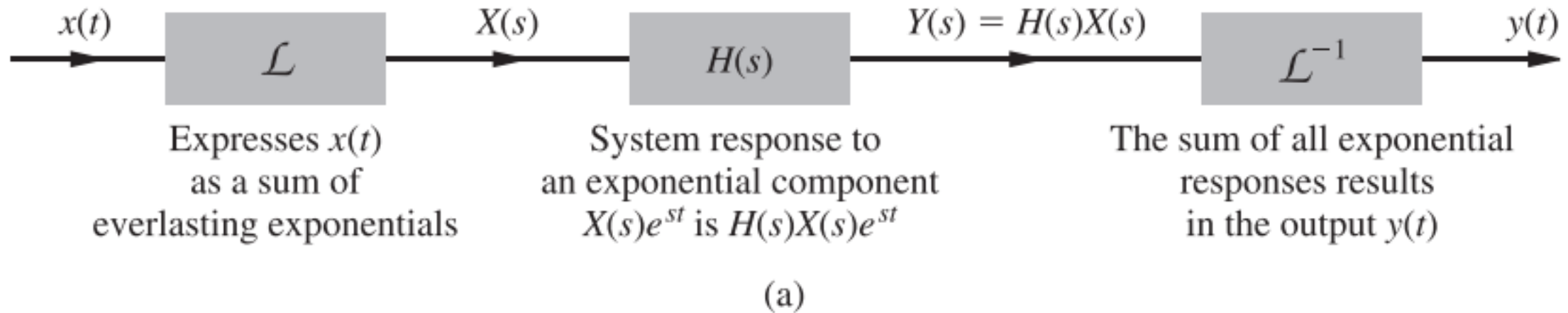


If $h(t)$ is the unit impulse response of an LTI system, then:

$$\begin{aligned} y(t) &= h(t) * e^{st} \\ &= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau \\ &= e^{st} H(s) \end{aligned}$$

Therefore, the output of an LTI system to a single everlasting exponential is the same as the input multiplied by the constant $H(s)$.

Frequency Domain analysis- LTIC Systems:





Frequency Domain analysis- LTIC Systems:

If $h(t)$ is the impulse response of an LTI system, then we have seen in previous lectures that the system's response to an input $x(t)$ is $y(t) = x(t) * h(t)$.

Using the time domain convolution property of the Laplace Transform:

$$Y(s) = X(s)H(s)$$

As the response $y(t)$ is the zero-state response of the LTI system to the input $x(t)$. Therefore, the transfer function of the system $H(s)$ is defined as:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{\mathcal{L}\{\text{zero-state response}\}}{\mathcal{L}\{\text{input}\}}$$



Frequency Domain analysis- LTIC Systems:

$$\begin{aligned} \frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \dots + a_{N-1} \frac{dy(t)}{dt} + a_N y(t) \\ = b_{N-M} \frac{d^M x(t)}{dt^M} + b_{N-M+1} \frac{d^{M-1} x(t)}{dt^{M-1}} + \dots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t) \end{aligned}$$

$$\mathcal{L} \left\{ \frac{d^k y(t)}{dt^k} \right\} = s^k Y(s) - (\text{initial conditions})$$

Assuming [zero initial conditions](#), important for transfer function calculation:

$$\begin{aligned} \frac{d^N y(t)}{dt^N} &\rightarrow s^N Y(s) \\ \frac{d^M x(t)}{dt^M} &\rightarrow s^M X(s) \end{aligned}$$

$$y(0^-) = \dot{y}(0^-) = \ddot{y}(0^-) = \dots = y^{(N-1)}(0^-) = 0$$

$$x(0^-) = \dot{x}(0^-) = \ddot{x}(0^-) = \dots = x^{(N-1)}(0^-) = 0$$

Frequency Domain analysis- LTIC Systems:

$$\begin{aligned} \frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \cdots + a_{N-1} \frac{dy(t)}{dt} + a_N y(t) \\ = b_{N-M} \frac{d^M x(t)}{dt^M} + b_{N-M+1} \frac{d^{M-1} x(t)}{dt^{M-1}} + \cdots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t) \end{aligned}$$

Left-hand side (output terms):

$$s^N Y(s) + a_1 s^{N-1} Y(s) + \cdots + a_N Y(s)$$

$$Y(s) [s^N + a_1 s^{N-1} + \cdots + a_N]$$

Right-hand side (input terms):

$$b_{N-M} s^M X(s) + b_{N-M+1} s^{M-1} X(s) + \cdots + b_N X(s)$$

$$X(s) [b_{N-M} s^M + b_{N-M+1} s^{M-1} + \cdots + b_N]$$



Frequency Domain analysis- LTIC Systems:

$$\begin{aligned}\frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \dots + a_{N-1} \frac{dy(t)}{dt} + a_N y(t) \\ = b_{N-M} \frac{d^M x(t)}{dt^M} + b_{N-M+1} \frac{d^{M-1} x(t)}{dt^{M-1}} + \dots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t)\end{aligned}$$

$$Y(s) [s^N + a_1 s^{N-1} + \dots + a_N] = X(s) [b_{N-M} s^M + \dots + b_N]$$

Differential equation (time domain) becomes an algebraic equation in the frequency domain (or S- domain)

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_{N-M} s^M + b_{N-M+1} s^{M-1} + \dots + b_N}{s^N + a_1 s^{N-1} + \dots + a_N}$$

Numerator polynomial: zeros

Denominator polynomial: poles



Frequency Domain analysis- LTIC Systems:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_{N-M}s^M + b_{N-M+1}s^{M-1} + \dots + b_N}{s^N + a_1s^{N-1} + \dots + a_N} \quad \longrightarrow \quad H(s) = \frac{P(s)}{Q(s)}$$

As any polynomial can be factorized in terms of its roots, then $H(s)$ becomes:

$$H(s) = K \frac{(s - z_1)(s - z_2) \cdots (s - z_M)}{(s - p_1)(s - p_2) \cdots (s - p_N)}$$

$$H(s) = K \frac{\prod_{i=1}^M (s - z_i)}{\prod_{i=1}^N (s - p_i)}$$

z_i : zero ----- cause $H(s) = 0$

p_i : poles ----- cause $H(s) = \infty$

Number of poles = N & Number of zeros = M

K : gain constant ($K = b_{N-M}$)



Frequency Domain analysis- LTIC Systems:

EXAMPLE 4.14 Laplace Transform to Find the Zero-State Response

Find the response $y(t)$ of an LTIC system described by the equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

if the input $x(t) = 3e^{-5t}u(t)$ and all the initial conditions are zero; that is, the system is in the zero state.

The system equation is

$$\underbrace{(D^2 + 5D + 6)}_{Q(D)}y(t) = \underbrace{(D + 1)}_{P(D)}x(t)$$

Therefore,

$$H(s) = \frac{P(s)}{Q(s)} = \frac{s + 1}{s^2 + 5s + 6}$$

Also,

$$X(s) = \mathcal{L}[3e^{-5t}u(t)] = \frac{3}{s + 5}$$

and

$$\begin{aligned} Y(s) &= X(s)H(s) = \frac{3(s + 1)}{(s + 5)(s^2 + 5s + 6)} \\ &= \frac{3(s + 1)}{(s + 5)(s + 2)(s + 3)} = \frac{-2}{s + 5} - \frac{1}{s + 2} + \frac{3}{s + 3} \end{aligned}$$

The inverse Laplace transform of this equation is

$$y(t) = (-2e^{-5t} - e^{-2t} + 3e^{-3t})u(t)$$

Frequency Domain analysis- LTIC Systems:

EXAMPLE 4.15 Laplace Transform to Find System Transfer Functions

Show that the transfer function of:

- (a) an ideal delay of T seconds is e^{-sT}
- (b) an ideal differentiator is s
- (c) an ideal integrator is $1/s$

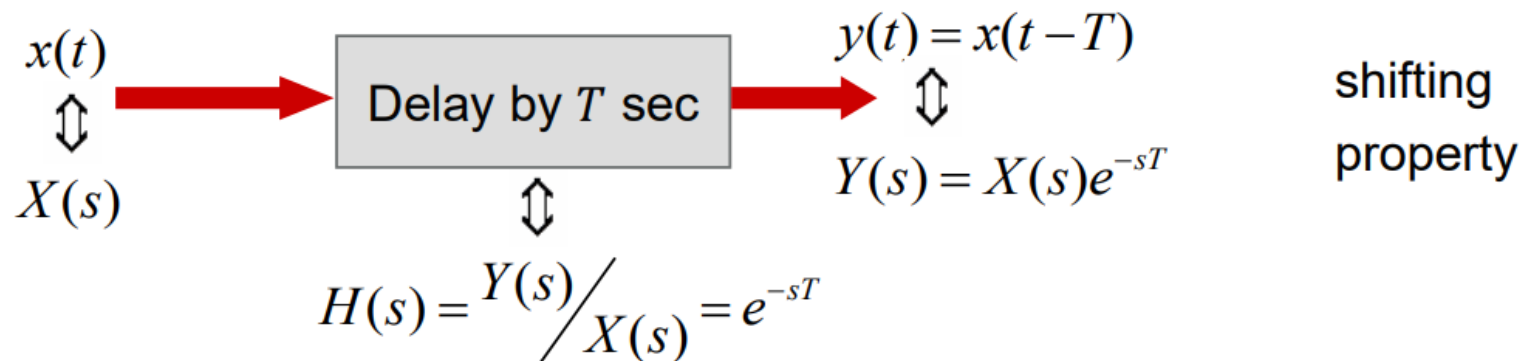
(a) **Ideal Delay.** For an ideal delay of T seconds, the input $x(t)$ and output $y(t)$ are related by

$$y(t) = x(t - T) \quad \text{and} \quad Y(s) = X(s)e^{-sT} \quad [\text{see Eq. (4.12)}]$$

Therefore,

$$H(s) = \frac{Y(s)}{X(s)} = e^{-sT} \quad (4.30)$$

System model- block to represent the delay T : $H(s) = e^{-sT}$



Frequency Domain analysis- LTIC Systems:

(b) Ideal Differentiator. For an ideal differentiator, the input $x(t)$ and the output $y(t)$ are related by

$$y(t) = \frac{dx(t)}{dt}$$

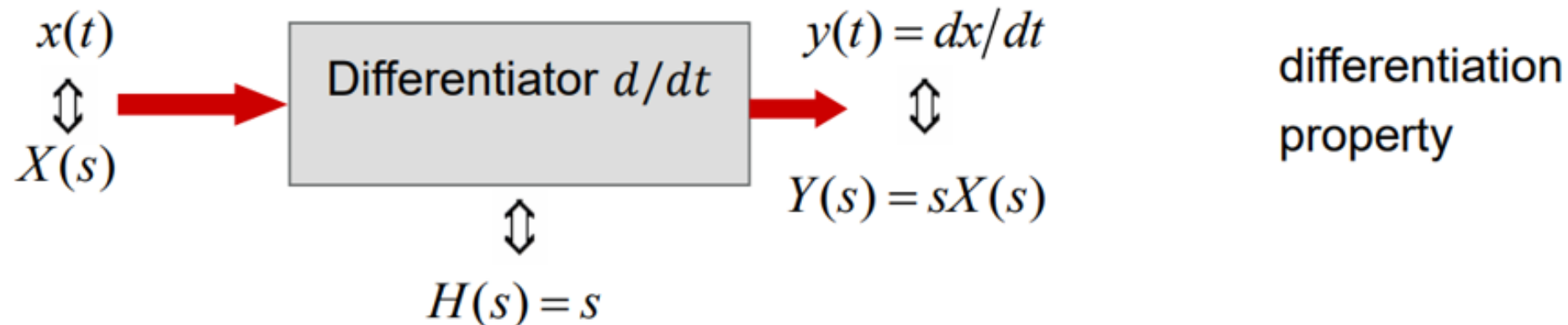
The Laplace transform of this equation yields

$$Y(s) = sX(s) \quad [x(0^-) = 0 \text{ for a causal signal}]$$

and

$$H(s) = \frac{Y(s)}{X(s)} = s \quad (4.31)$$

System model- block to represent the Differentiator: $H(s) = S$



Frequency Domain analysis- LTIC Systems:

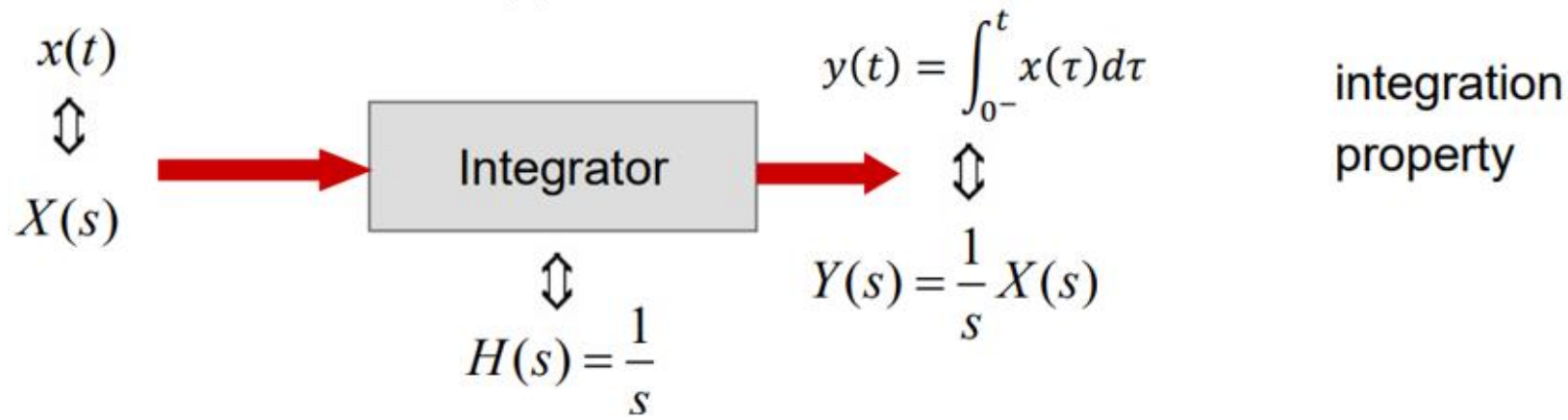
(c) **Ideal Integrator.** For an ideal integrator with zero initial state, that is, $y(0^-) = 0$,

$$y(t) = \int_0^t x(\tau) d\tau \quad \text{and} \quad Y(s) = \frac{1}{s}X(s)$$

Therefore,

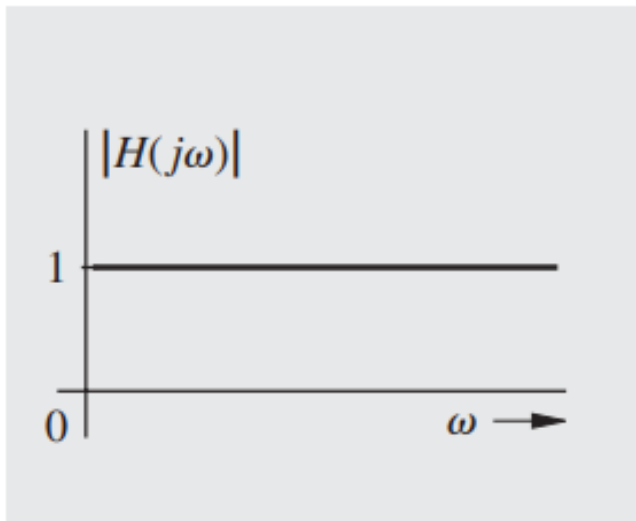
$$H(s) = \frac{1}{s} \quad (4.32)$$

System model- block to represent the Integrator: $H(s) = 1/S$



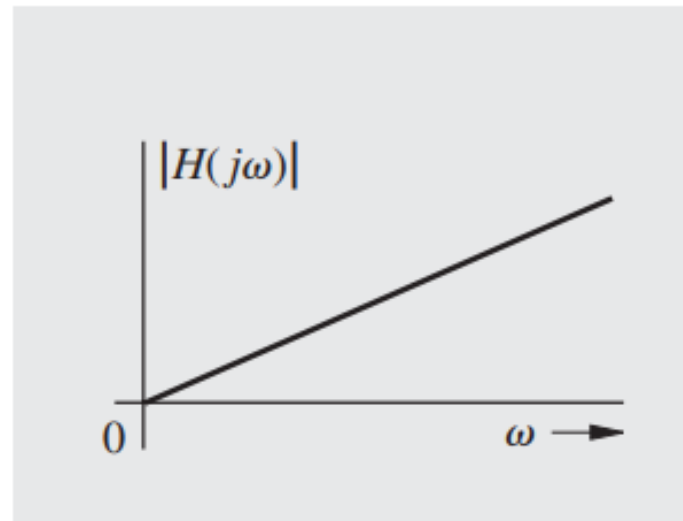
Frequency Domain analysis- LTIC Systems:

Ideal Delay



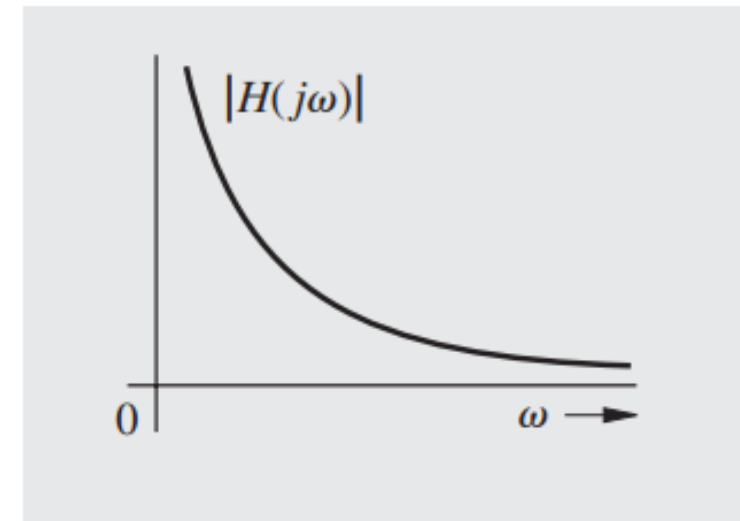
All Pass Filter

Ideal Differentiator



High Pass Filter

Ideal Integrator



Low Pass Filter

Frequency Domain analysis- LTIC Systems:

DRILL 4.8 Differential Equation and Zero-State Response from a System Transfer Function

For an LTIC system with transfer function,

$$H(s) = \frac{s+5}{s^2+4s+3}$$

- (a) Describe the differential equation relating the input $x(t)$ and output $y(t)$.
- (b) Find the system response $y(t)$ to the input $x(t) = e^{-2t}u(t)$ if the system is initially in zero state.

ANSWERS

(a) $\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 5x(t)$

(b) $y(t) = (2e^{-t} - 3e^{-2t} + e^{-3t})u(t)$

CT System Analysis
 $H(s)$: Poles & Zeros
System Stability Check



Frequency Domain analysis- LTIC Systems:

LTIC BIBO Stability Check @ Frequency Domain (S-plane)

$$H(s) = K \frac{(s - z_1)(s - z_2) \cdots (s - z_M)}{(s - p_1)(s - p_2) \cdots (s - p_N)} \quad \longrightarrow \quad H(s) = \frac{P(s)}{Q(s)}$$

z_i : zero ----- cause $H(s) = 0$

p_i : poles ----- cause $H(s) = \infty$

Number of poles = N & Number of zeros = M

K : gain constant ($K = b_{N-M}$)

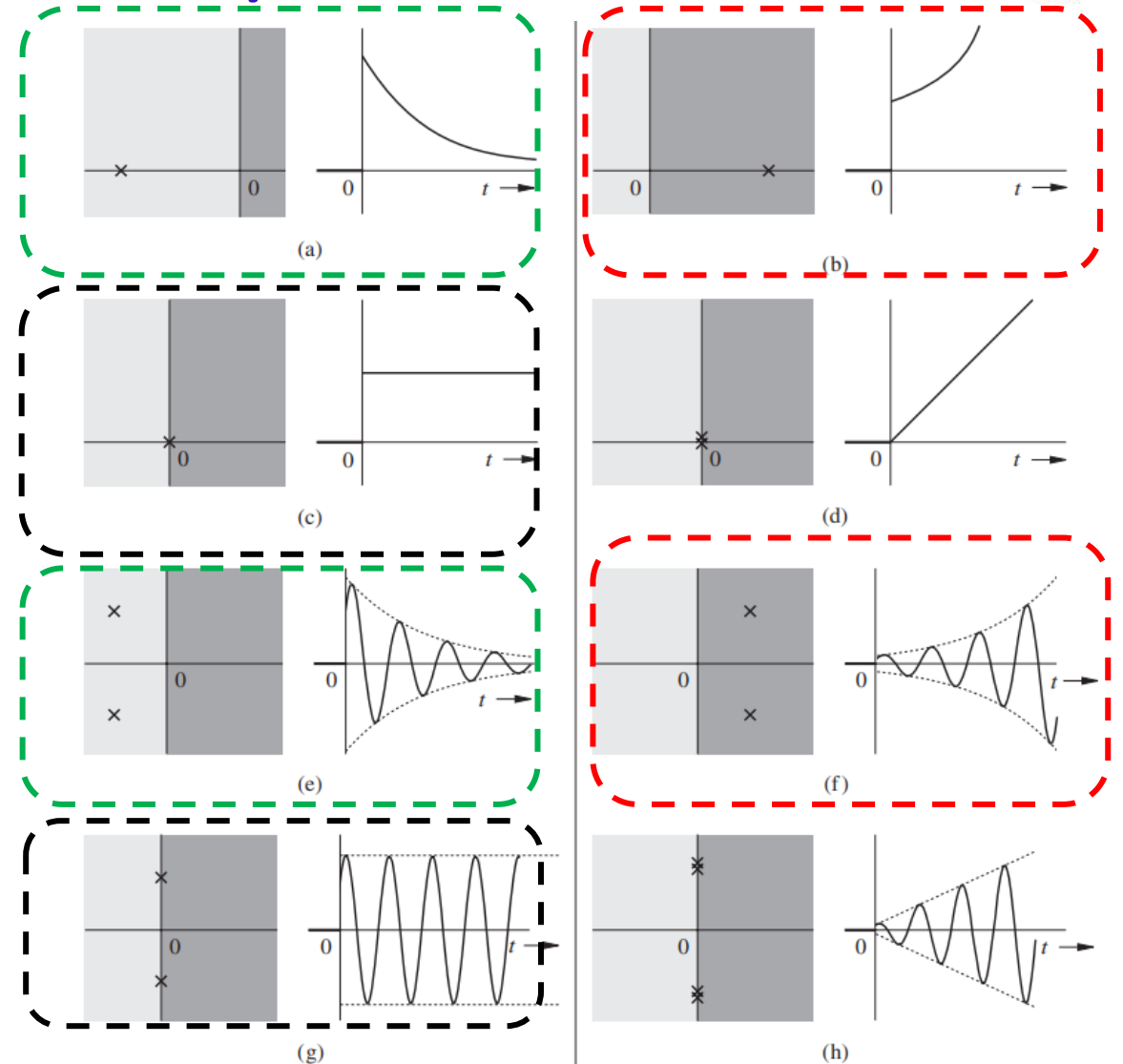
Frequency Domain analysis- LTIC Systems:

LTIC BIBO Stability Check

In the time domain: $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

In the Frequency domain @ S- plane:

All poles of $H(s)$ must lie in the left-half of the complex plane (LHP). Poles in the right-half plane (RHP) or on the imaginary axis may lead to instability.



Frequency Domain analysis- LTIC Systems:

LTIC BIBO Stability Check

$$H(s) = \frac{1}{s+3} + \frac{2}{s+4}$$

Example: 1

The poles are at $s=-3$ and $s=-4$.

Freq. domain check: For BIBO stability, all poles must be in the left half-plane (LHP). Here, both lie in the LHP, so the given system is BIBO stable.

Time domain check:

$$h(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s+3} + \frac{2}{s+4} \right\} = e^{-3t}u(t) + 2e^{-4t}u(t)$$

Both components are converging [$e^{-at}u(t)$], the system is BIBO stable as $\int_0^{\infty} |h(t)|dt < \infty$

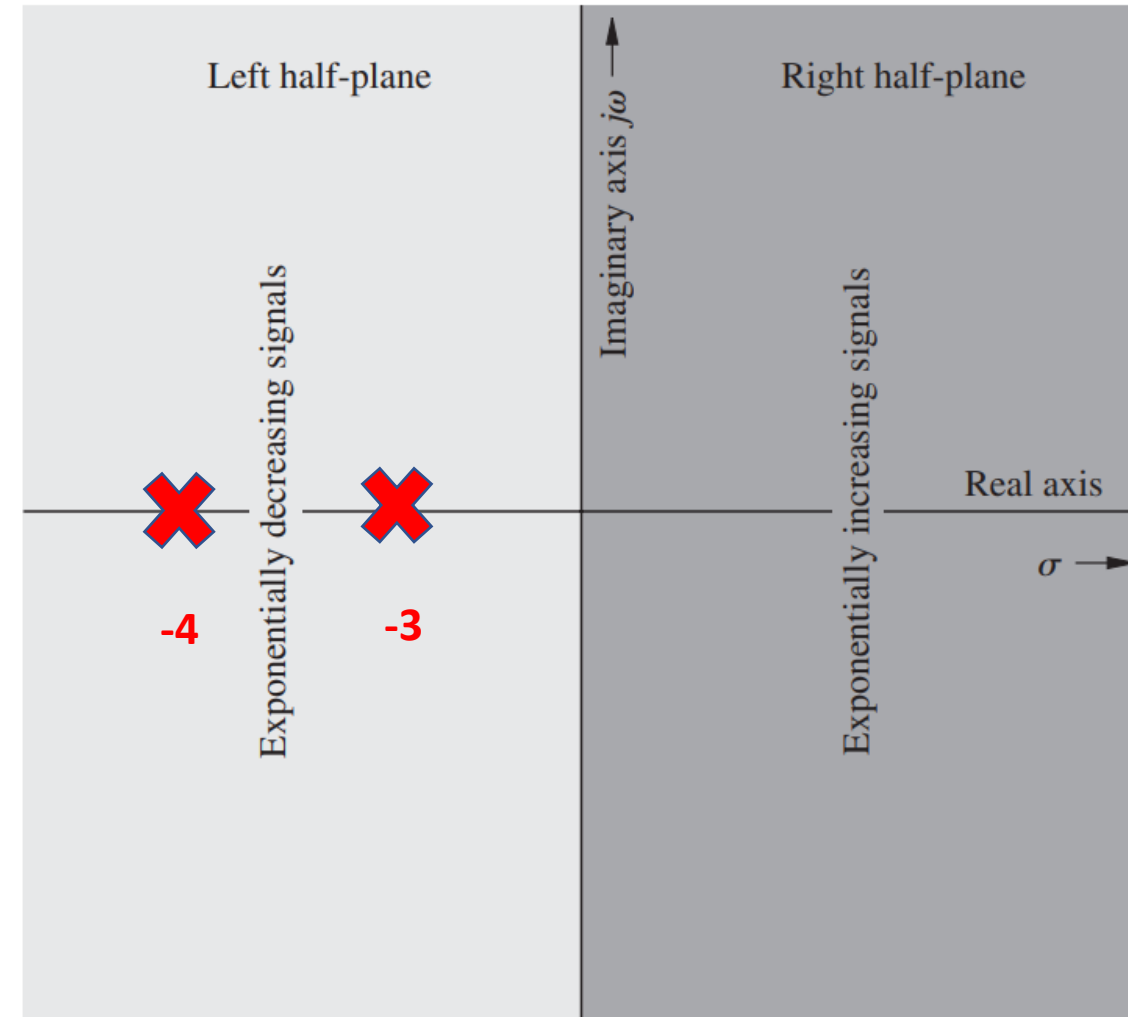


Figure 1.22 Complex frequency plane.

Frequency Domain analysis- LTIC Systems:

LTIC BIBO Stability Check

$$H(s) = \frac{1}{s+3} + \frac{2}{s-4}$$

Example: 2

The poles are at $s=-3$ and $s=4$.

Freq. domain check: For BIBO stability, all poles must be in the left half-plane (LHP). Here, one pole in the RHP, so the given system is NOT BIBO stable.

Time domain check:

$$h(t) = e^{-3t}u(t) + 2e^{4t}u(t)$$

$$\int_0^{\infty} |e^{-3t}| dt = \frac{1}{3}$$

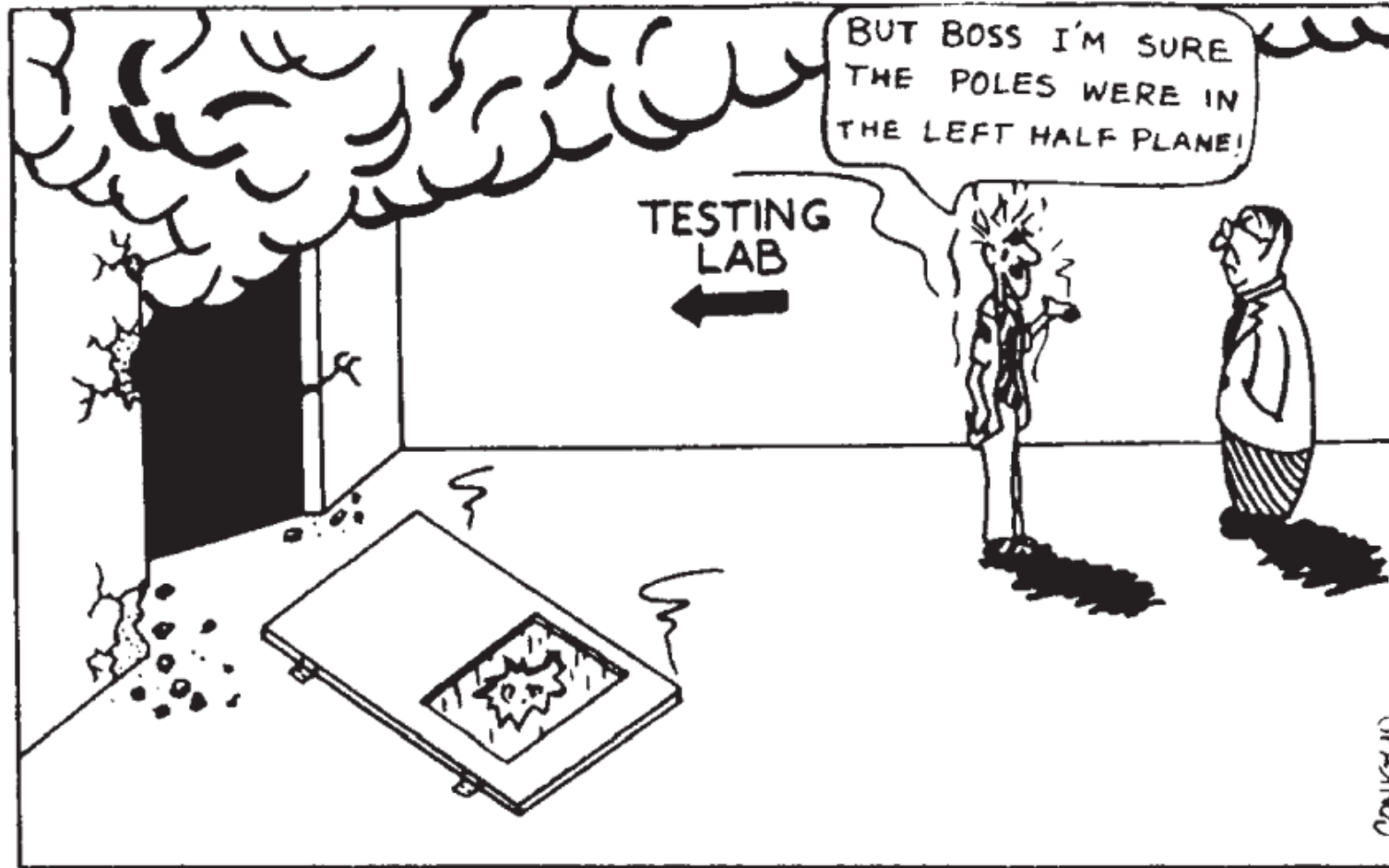
$$\int_0^{\infty} |2e^{4t}| dt = \infty$$

Since the second term grows exponentially, the integral diverges, confirming time-domain instability.



Figure 1.22 Complex frequency plane.

Frequency Domain analysis- LTIC Systems:



Beware of right half-plane poles!

End of Session # 02
End of Laplace Transform

Next: Fourier Transform
(The last topic of the course)

Thanks!